

Exploración robótica

Cátedra: Introducción a la robótica móvil



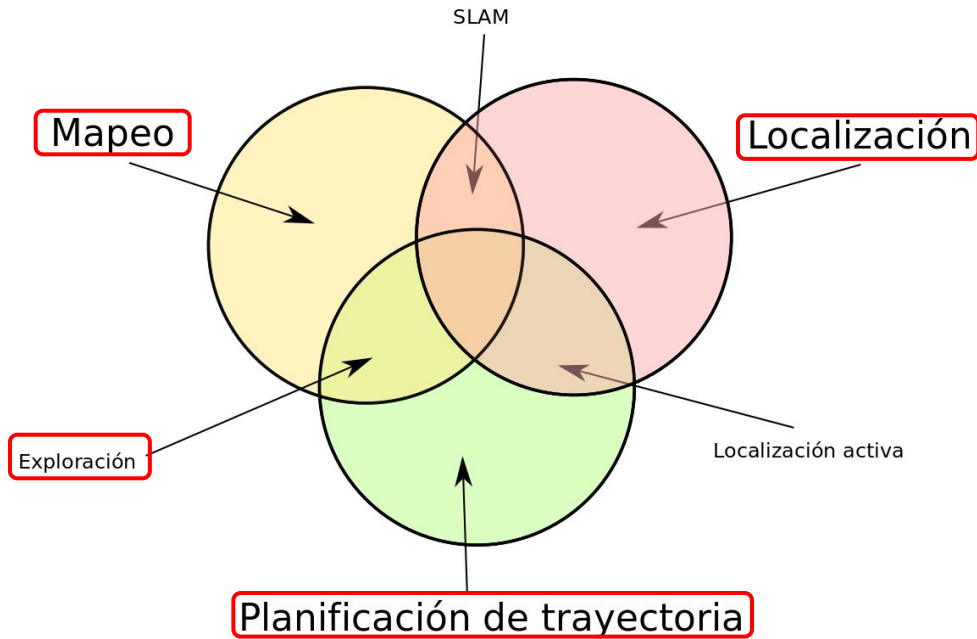
Martin Nieves

Centro de Investigación en Informática para la Ingeniería
Universidad Tecnológica Nacional - Facultad Regional Córdoba.

Agenda

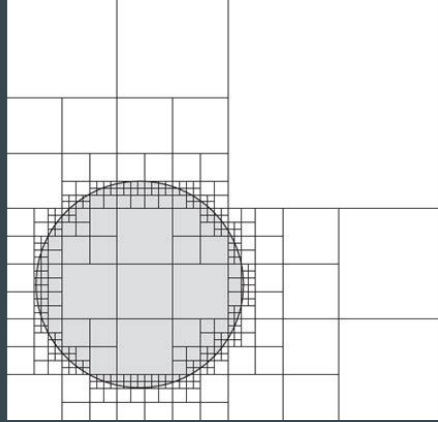
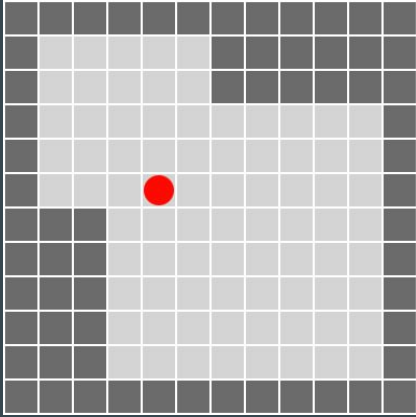
- Concepto de exploración
- Mapas
- Ambientes no estructurados
- Panorama actual
- Ejemplos

Exploración?

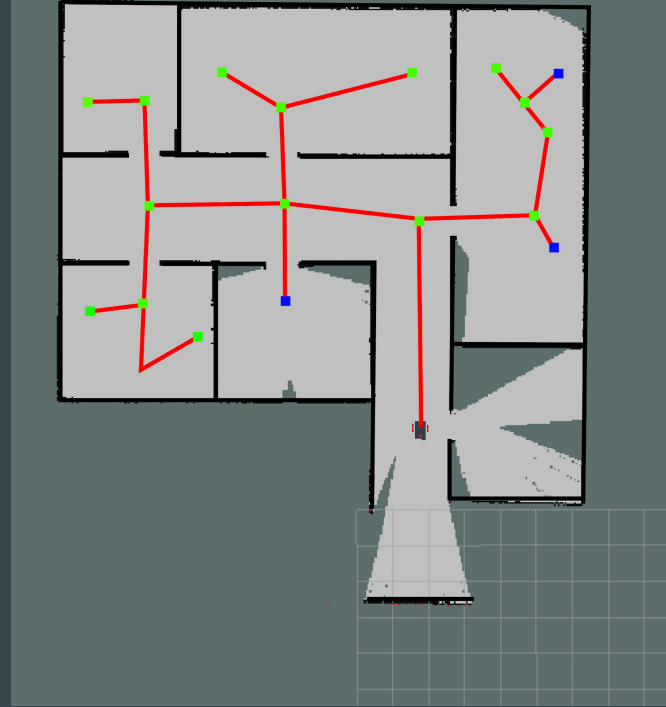
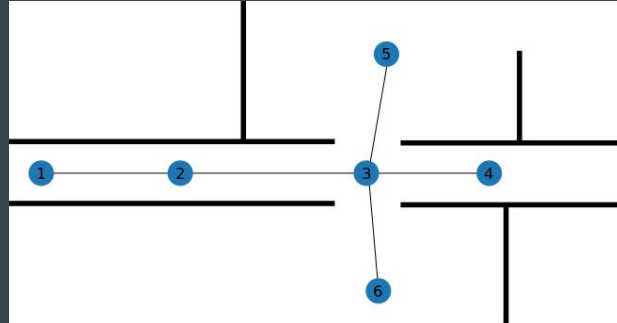


Mapas?

Métricos



Topológicos



Ambientes estructurados



A photograph of a dark, rocky tunnel. The walls are uneven and textured, with some light reflecting off the surfaces. A bright light source is visible at the far end of the tunnel, creating a strong glow and casting long shadows. The floor of the tunnel is covered in rocks and debris.

HOW DEEP DOES IT GO?



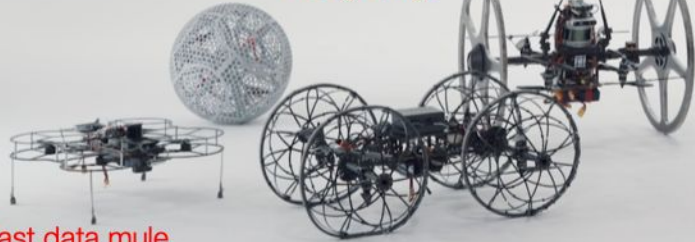
DARPA Subt Challenge



URBAN CIRCUIT

DARPA Subt Challenge

Hybrid ground/aerial
Long-range



Fast data mule

Large climbs/slopes



Fast data mule



Repeater node dropper

Vertical shafts





DARPA SUBTERRANEAN CHALLENGE



Evan Ackerman
@BotJunkie

I don't know whether it was on purpose or not but this is the slickest left turn I've ever seen [@nctu_t](#)
[@DARPA #SubTChallenge](#)



6:01 PM · 8/18/19 · [TweetDeck](#)

5 Retweets 15 Likes



Martin Nieves
@minievas

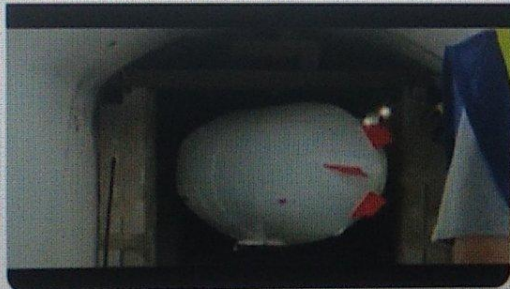
Wow! [#SubTChallenge](#)
In dirigible entrando en una mina

[Translate Tweet](#)



Evan Ackerman @BotJunkie · 4d

It's happening!!! THE BLIMP IS ENTERING THE MINE, I repeat, THE BLIMP IS ENTERING THE MINE [@nctu_t](#) [@DARPA #SubTChallenge](#)



11:02 PM · 8/17/19 · [Twitter for Android](#)

1 Like



Shreyas Skandan
@ShreyasSkandan

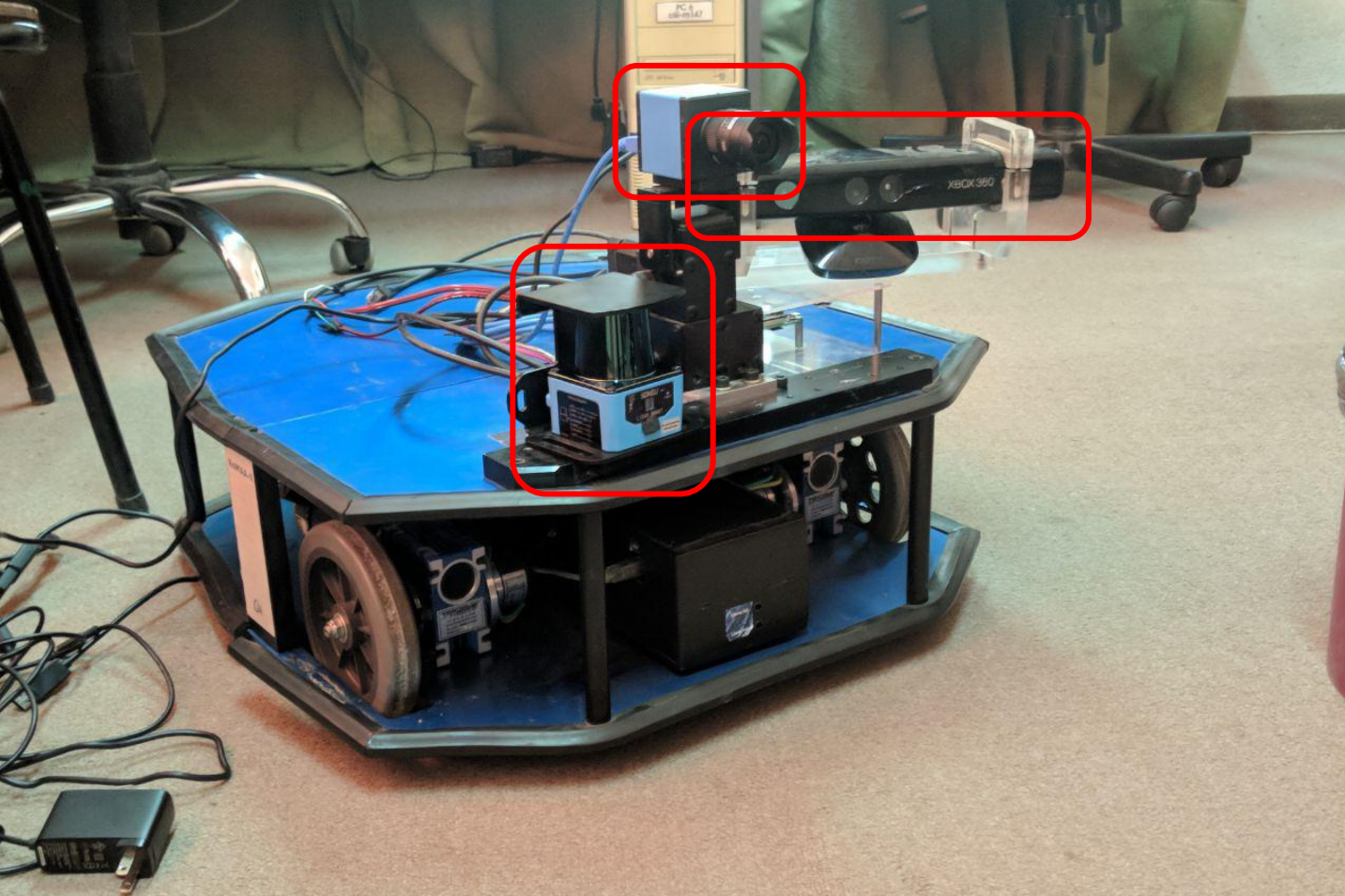
Find someone who looks at you the way [@BotJunkie](#) looks at Blimps.
[@DARPA #SubTChallenge](#)



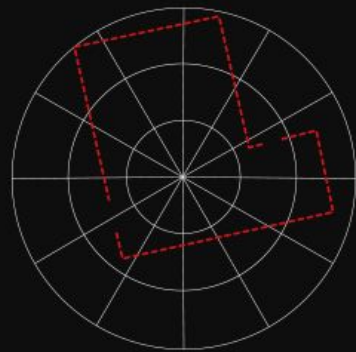
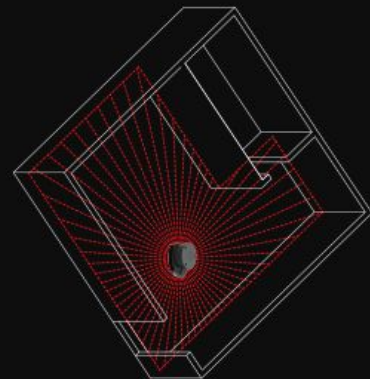
Eyan Ackerman

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12 Likes



ROMAA-II



Simuladores



gitlab.com/martinnievas/ros-jaiio-2019



GAZEBO

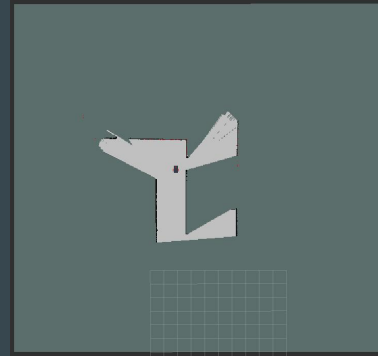
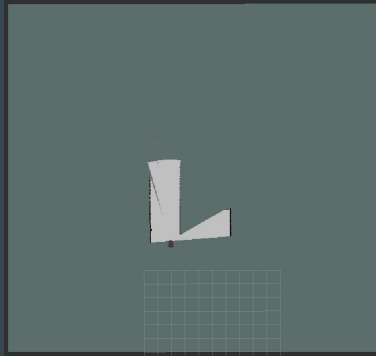


Código y
simulaciones
disponibles

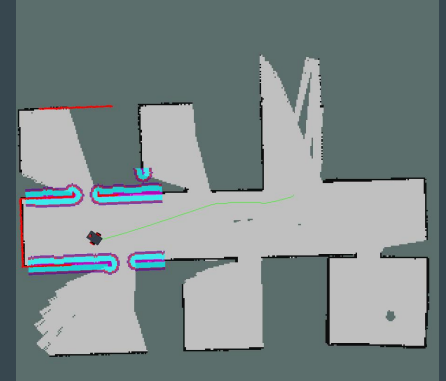
Ejemplos

gmapping

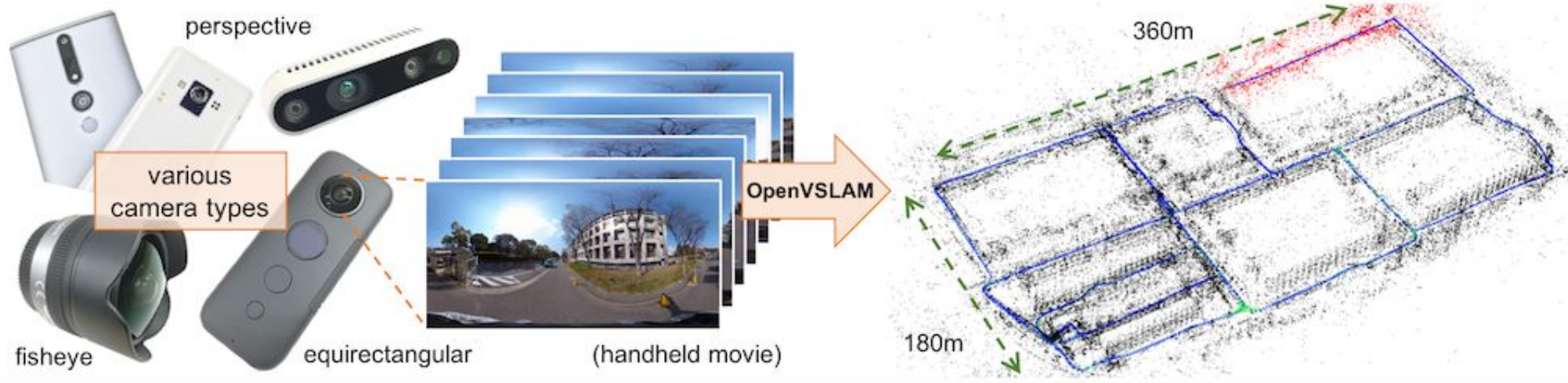
frontier_exploration



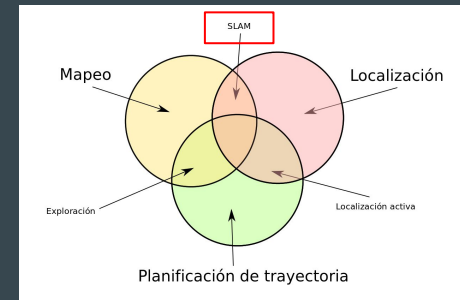
Los programas utilizados están aquí -> <https://gitlab.com/martinnievas/ros-jaiio-2019>



SLAM - simultaneous localization and mapping



OpenVSLAM (Sumikura, Shinya et. al)
<https://github.com/xdspacelab/openvslam>



OpenVSLAM - Ejemplo en drones



chakofpv [Seguir](#)

158 publicaciones 2,332 seguidores 601 seguidos

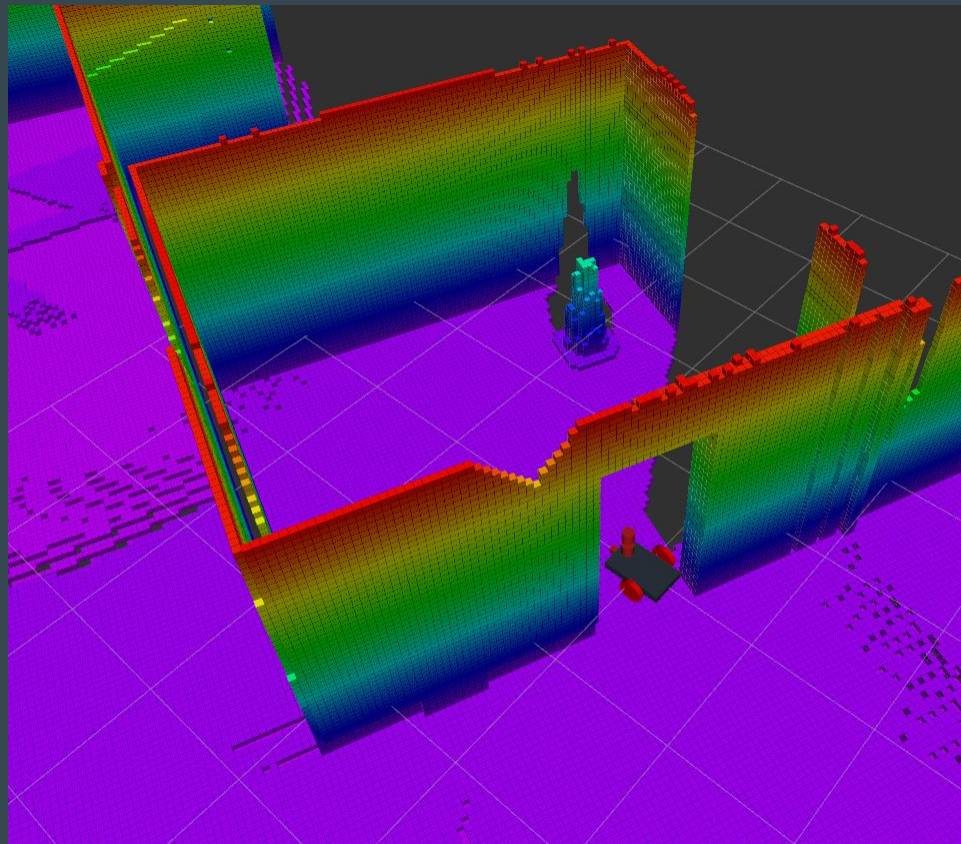
ChaKo FPV | Drone Pilot
Content creator 🤖🚁🇨🇦
Cordoba, Argentina 🇨🇦👉
Cinewhoop? 🇺🇸 -> stanfpv.com
FPV Gear? 🇨🇦 -> rcjuampa.com.ar
DJI? 🇨🇦 -> rconline.com.ar
Umma95X Action!:
youtu.be/BXwUN3SU9ml

- Utilizar un video de drone como entrada
(Gracias a chako del IUA por el video)
- Generar un mapa como salida mientras
localizamos la cámara en el mismo



Mapas en 3D

Ejemplo octomap



No todo es color de rosas

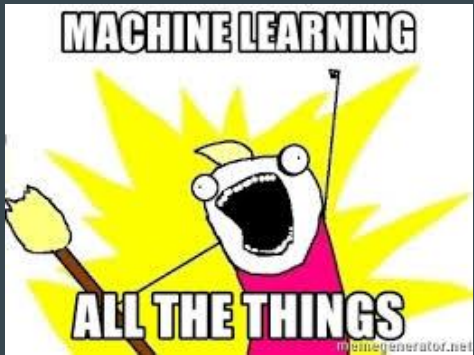
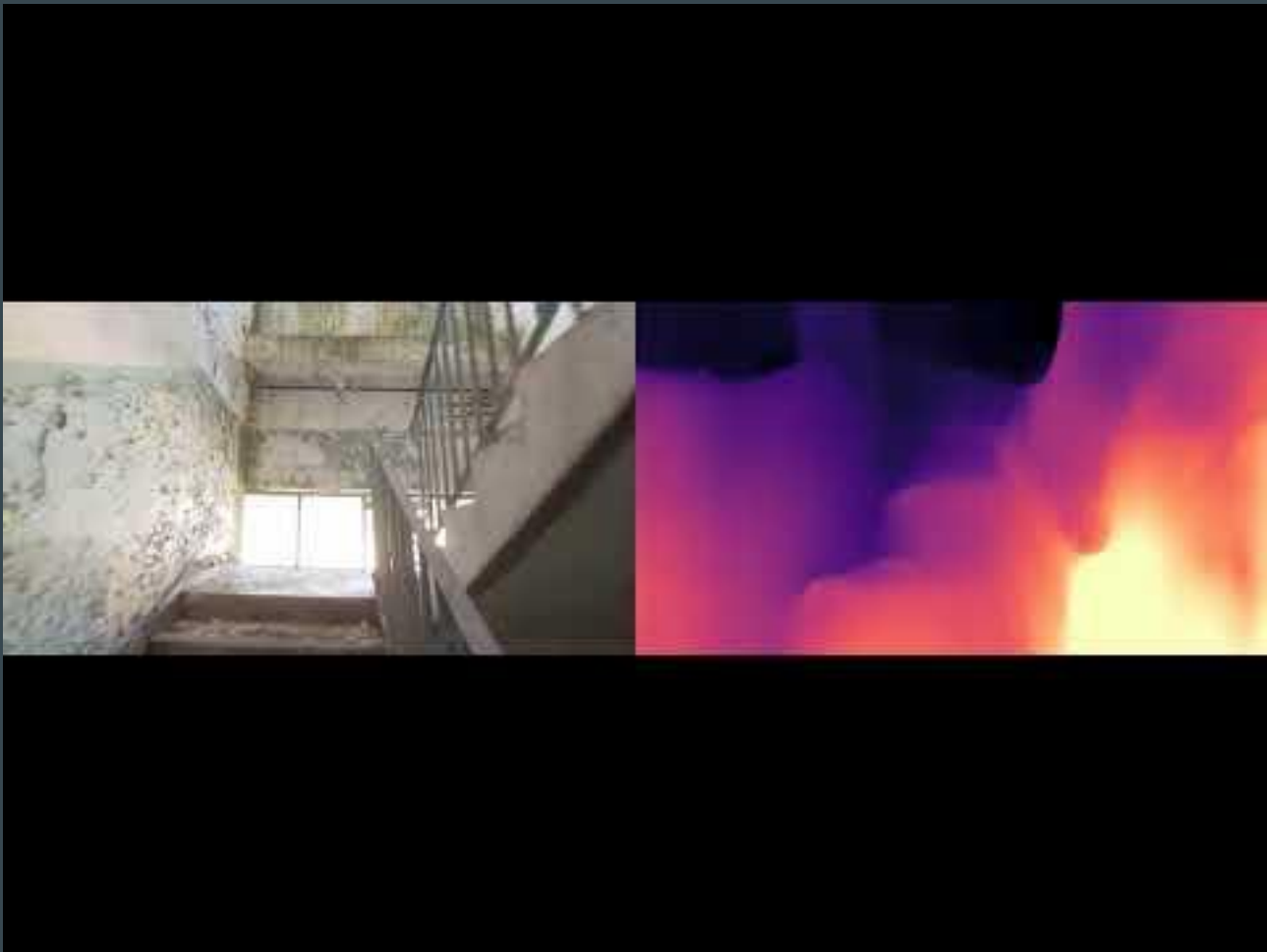
- Ruido de los sensores
- Disponibilidad de cámaras
- Robots

Cámaras RGB-D realsense

realsense-viewer

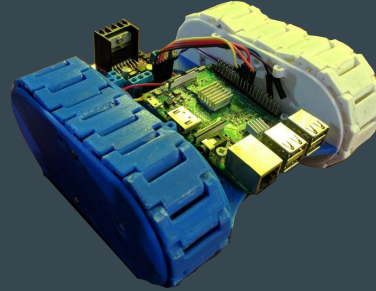
ejemplo rtabmap





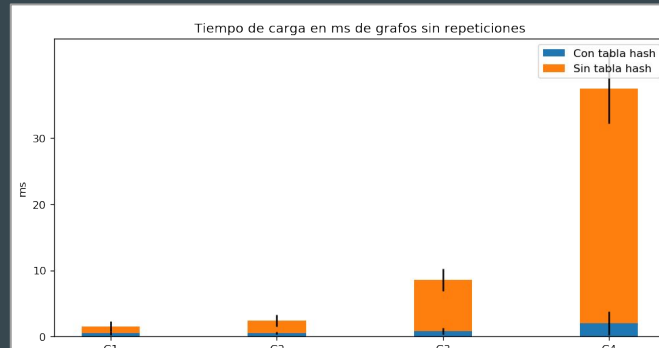
Exploración eficiente de ambientes no estructurados con robot heterogéneos

- Utilizar múltiples robots
- Reducir el consumo de energía
- Plataformas más eficientes
- Considerar la batería durante la exploración
- Robots más “económicos”



Grafo	Con Tabla Tiempo[ms]	Sin Tabla Tiempo[ms]
G1[63]	0.49 (± 0.29)	1.03 (± 0.74)
G2[127]	0.51 (± 0.17)	1.88 (± 0.88)
G3[252]	0.81 (± 0.47)	7.76 (± 1.69)
G5[509]	1.99 (± 1.76)	35.50 (± 5.33)
G9[1021]	2.08 (± 1.37)	102.73 (± 8.83)

Figura 4: Tiempos promedios de exploración



¿Preguntas?

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martinnievas.com



Recursos de robótica: blog.martinnievas.com